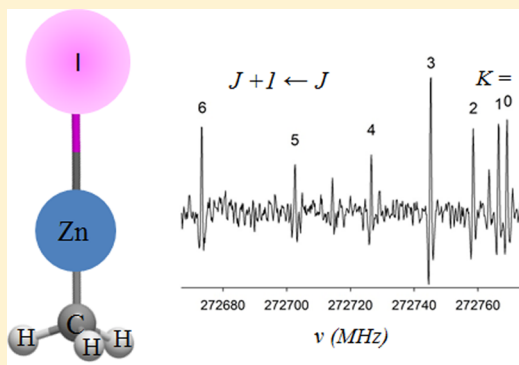


Structural Determination and Gas-Phase Synthesis of Monomeric, Unsolvated IZnCH_3 ($\tilde{X}^1A_1$): A Model Organozinc HalideMatthew P. Bucchino,[†] Justin P. Young,[‡] Phillip M. Sheridan,^{*,‡} and Lucy M. Ziurys[†][†]Department of Chemistry, Department of Astronomy, and Steward Observatory, University of Arizona, 933 North Cherry Avenue, Tucson, Arizona 85721, United States[‡]Department of Chemistry and Biochemistry, Canisius College, 2001 Main Street, Buffalo, New York 14208, United States

S Supporting Information

ABSTRACT: The first experimental structure of a monomeric organozinc halide, IZnCH_3 , has been measured using millimeter-wave direct absorption spectroscopy in the frequency range 256–293 GHz. IZnCH_3 is a model compound for organozinc halides, widely used in cross-coupling reactions. The species was produced in the gas phase by reaction of zinc vapor with iodomethane in the presence of a dc discharge. IZnCH_3 was identified on the basis of its pure rotational spectrum as well as those of the isotopically substituted species $^{66}\text{ZnCH}_3$, $^{64}\text{Zn}^{13}\text{CH}_3$, and $^{64}\text{ZnCD}_3$. IZnCH_3 is unmistakably a symmetric top molecule ($\tilde{X}^1A_1$) belonging to the C_{3v} point group, in agreement with DFT calculations, with the following experimentally determined structural parameters: $r_{\text{IZn}} = 2.4076(2)$ Å, $r_{\text{ZnC}} = 1.9201(2)$ Å, $r_{\text{CH}} = 1.105(9)$ Å, and $\angle \text{H-C-H} = 108.7(5)^\circ$. The basic methyl group geometry is not significantly altered in this molecule. Experimental observations suggest that IZnCH_3 is synthesized in the gas phase by direct insertion of activated atomic zinc into the carbon–iodine bond of iodomethane.

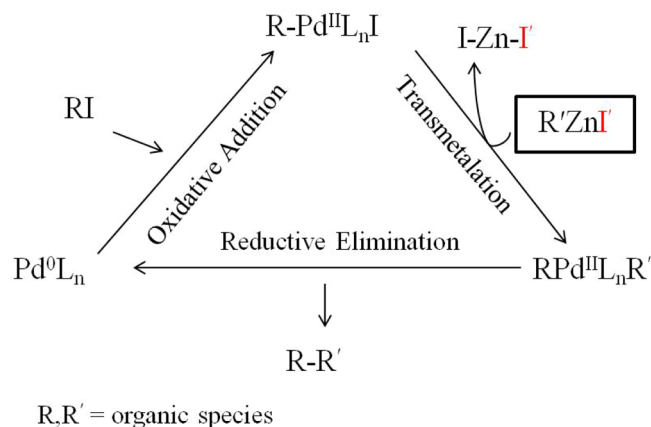


■ INTRODUCTION

Organozinc compounds are known to be a very useful group of reagents for organic synthesis.^{1–5} Organozinc halides, RZnX , are one major class of such compounds. These species are characterized by their ease of preparation, high functional group compatibility, and excellent reactivity, which in some cases requires an appropriate catalyst. They are typically synthesized by oxidative addition of zinc to an organic halide (usually a bromide or iodide) in solution and can be prepared in high yield under mild reaction conditions. They are particularly appealing as the alkyl nucleophile in C–C bond formation reactions—transformations that are crucial in synthesizing complex structures. For example, the Fukuyama coupling reaction is used to create more complex ketones by reaction of RZnX with a thioester in the presence of a palladium catalyst.⁶ Of particular recent interest is the Negishi cross-coupling reaction, where addition of $\text{R}'\text{ZnX}$ to an organic halide RX in the presence of a palladium or nickel catalyst yields $\text{R}'\text{—R}$ species.^{7,8}

Although widely used in organic synthesis, the mechanistic details of the Negishi reaction are not completely understood.⁹ The process occurs via three basic steps, as shown in Scheme 1.¹⁰ The first step is oxidative addition of an organohalide (RI) to a Pd^0 catalyst, followed by transmetalation, in this case, of an organozinc halide ($\text{R}'\text{ZnI}'$) with the species ($\text{R—Pd}^{\text{II}}\text{L}_n\text{I}$) to produce $\text{R—Pd}^{\text{II}}\text{L}_n\text{R}'$. Reductive elimination of this intermediate then yields the product $\text{R—R}'$ as well as regenerating the Pd^0 catalyst. Kinetics studies conducted utilizing ^{19}F and ^{31}P

Scheme 1



NMR in tetrahydrofuran (THF) support this general mechanism^{9,11} but also reveal the complex nature of cis–trans isomerization during the transmetalation step. Theoretical work suggests that the transformation is even more complicated. For example, DFT calculations show that the transmetalation stage may involve bimetallic intermediate complexes.¹² Examination of the potential energy surfaces of

Received: August 5, 2014

Revised: October 24, 2014

Published: November 12, 2014

Ni-catalyzed alkyl–alkyl couplings shows that the oxidative addition occurs via a radical pathway involving transfer of the halogen atom followed by alkyl radical attack.¹³ Mechanistic results appear to vary with the systems investigated, including the reactants and solvents. Clearly elucidation of the complex pathways involved in the Negishi reaction is not trivial.

Methylzinc halides CH_3ZnX have been used as model compounds to investigate the functional group sensitivity and nucleophilic character of RZnX species;¹⁴ they have also been employed to evaluate potential energy surfaces and energetics of the Negishi reaction and related cross-coupling processes.¹³ However, few quantitative physical properties of methylzinc halides, such as bond lengths and angles, have been experimentally measured. For CH_3ZnI , thus far only low-resolution infrared spectra have been recorded, in solution with either THF or dimethoxyethane (DME) solvents,¹⁵ and a proton NMR spectrum has been recorded in a temperature-dependent Schlenk equilibrium investigation.¹⁶ The structures of more complex alkylzinc species have been investigated. For example, an X-ray crystal structure of ethylzinc iodide has been reported, revealing a unit cell with four EtZnI molecules.¹⁷ Various dimethylformamide-solvated alkylzinc cations have also been studied in the gas phase using electrospray ionization mass spectrometry, infrared multiphoton dissociation spectroscopy, and DFT calculations.^{18–21}

Here we report the first observation of a monomeric, unsolvated CH_3ZnX species, (IZnCH_3) ($\bar{X}^1\text{A}_1$). Gas-phase direct absorption millimeter-wave methods have been used to measure the pure rotational spectra of this molecule. Observation of $\text{I}^{64}\text{ZnCH}_3$ and three other isotopologues has established an unambiguous identification of this molecule and enabled a precise determination of its structure. DFT calculations have also been carried out for IZnCH_3 . Structural parameter comparisons with similar species also studied by pure rotational spectroscopy have provided insight into the bonding in IZnCH_3 . Additionally, experimental reaction conditions suggest that IZnCH_3 was synthesized in the gas phase by a zinc insertion mechanism as opposed to forming by recombination of radical fragments generated in a discharge.

■ EXPERIMENTAL AND COMPUTATIONAL METHODS

The IZnCH_3 molecule was characterized in the gas phase by pure rotational spectroscopy using one of the millimeter-wave direct absorption spectrometers of the Ziurys group.²² Briefly, the spectrometer consists of a tunable, coherent source of mm/submm radiation, a water-cooled, vacuum-sealed, stainless steel reaction cell, and a detector. The radiation sources are Gunn oscillator/varactor multiplier combinations, which span a range of 65–850 GHz; the source is phase locked to a stable frequency reference. The 0.5 m long reaction cell (diameter 0.1 m), which contains a Broida-type oven for metal vaporization, is evacuated to ~ 1 mTorr using a Roots blower pumping system. The detector is a helium-cooled InSb hot-electron bolometer. Molecules of interest are continuously synthesized in the cell in the gas phase. Radiation from the source is scanned in frequency and quasi-optically propagated through the cell and into the detector via a scalar feed horn, a wire polarizing grid, a rooftop mirror, and a series of Teflon lenses. When a given frequency is resonant with a rotational transition, an absorption signal is measured. Data is collected in scans 110 MHz in width, which are successively added to generate broad-band spectral coverage. In order to eliminate

background noise, phase-sensitive detection is carried out by frequency modulation of the Gunn oscillator. Signals from the InSb detector are processed by a lock-in amplifier at twice the frequency modulation rate; therefore, absorption signals have a second-derivative line profile.

IZnCH_3 was created in the gas phase in a dc discharge plasma by reaction of zinc vapor and CH_3I vapor in the presence of argon. Zinc metal pieces (99.9% purity, Sigma-Aldrich) were resistively heated in a Broida-type oven attached to the reaction cell to generate the metal vapor. About 4 mTorr of CH_3I vapor (Sigma-Aldrich) was added to the cell through a tube located directly above the oven, while 15 mTorr of argon was supplied from underneath the oven to act as a carrier gas. Argon was also flowed over the two Teflon lenses that seal the reaction cell in order to minimize condensation of zinc metal, which attenuates millimeter waves. Optimal discharge conditions were 0.060 A at 460 V, which generated a pale purple plasma in the gas mixture, likely due to atomic emission from zinc. $\text{IZn}^{13}\text{CH}_3$ and IZnCD_3 were synthesized under similar conditions using I^{13}CH_3 (99% purity, Sigma-Aldrich) and ICD_3 (99.5% purity, Sigma-Aldrich) as the respective precursors. Zinc isotopologues were observed in their natural abundance (^{64}Zn : ^{66}Zn , 48.6%:27.9%).

Precise transition frequencies were measured using an averaged pair of 5 MHz wide scans, one scan of the pair recorded in increasing frequency and the other scan obtained in decreasing frequency. Spectral lines were fit with Gaussian profiles in order to determine the center frequency. Rotational transitions from $\text{I}^{64}\text{ZnCH}_3$ were sufficiently intense such that only one scan pair was necessary; those of $\text{I}^{64}\text{ZnCH}_3$ ($\nu_4 = 1$), $\text{I}^{66}\text{ZnCH}_3$, $\text{I}^{64}\text{ZnCD}_3$, and $\text{I}^{64}\text{Zn}^{13}\text{CH}_3$ were much weaker and required averages of as many as five pairs. Typical line widths varied from 500 to 700 kHz in the frequency range of 256–293 GHz. Experimental uncertainty is approximately ± 50 kHz.

All geometric optimization and harmonic vibrational frequency calculations were performed using Gaussian 09 (ref 23). DFT calculations utilized either the B3LYP^{24,25} or the B3PW91²⁶ functionals. The 6-311G++(3df,2p) basis set was used for the H, C, and Zn atoms, while either the LanL2DZ (and LanL2DZ effective core potential, ECP)²⁷ or the DGDZVP²⁸ basis sets were used for the iodine atom.

■ RESULTS AND ANALYSIS

The 272–278 GHz frequency range was continuously scanned while reacting zinc with CH_3I in the presence of a dc discharge, initially in search of pure rotational transitions of ZnCH_3 ($\bar{X}^2\text{A}_1$). Three weak and harmonically related transitions, each exhibiting a symmetric top pattern characteristic of a molecule with a C_3 axis of symmetry, were also identified among numerous strong lines attributed to CH_3I . Removal of the zinc vapor caused these weak spectral features to promptly disappear; therefore, they could be attributed to a zinc-bearing species. However, when CH_3I was replaced with $(\text{CH}_3)_4\text{Sn}$, a methyl group donor used successfully to synthesize other MCH_3 molecules in the Ziurys lab,^{29–31} these molecular signals vanished, which suggested that they originated from an iodine-containing species. A preliminary value of the B rotational constant for these harmonically related lines was ~ 1.1 GHz, far lower than the experimental value of ~ 9.2 GHz reported for ZnCH_3 from optical spectroscopic studies³² and consistent with the presence of a heavy atom such as iodine. The harmonically related lines did not exhibit a spin-rotation fine structure splitting, as would be the case for ZnCH_3 . Therefore,

these transitions could not be due to ZnCH_3 . $\text{I}^{64}\text{ZnCH}_3$ with a linear $\text{I}-\text{Zn}-\text{C}$ backbone (i.e., a C_3 axis) seemed to be a logical molecular source of these lines. A search for $\text{I}^{66}\text{ZnCH}_3$ was then conducted, and spectral signatures were found at the expected frequency shift for the zinc change in mass. Final confirmation of the identification was obtained through synthesis of $\text{I}^{64}\text{ZnCD}_3$ and $\text{I}^{64}\text{Zn}^{13}\text{CH}_3$ and measurement of their rotational transitions as well as via DFT calculations.

Figure 1 shows the spectrum of the $J = 116 \leftarrow 115$ rotational transition of $\text{I}^{64}\text{ZnCH}_3$ recorded near 273 GHz. The pattern of

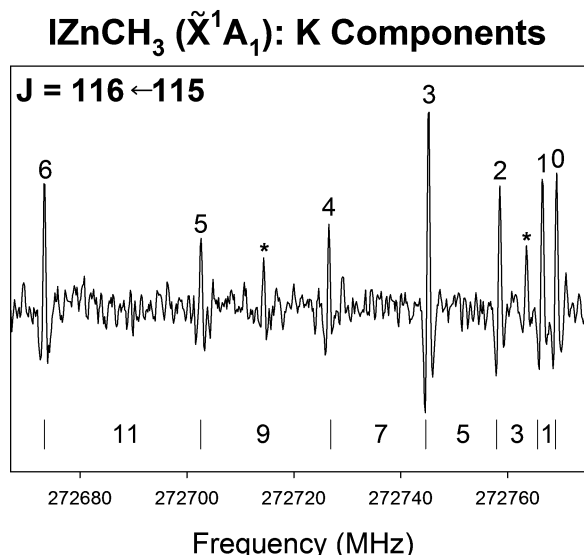


Figure 1. Spectrum of $\text{I}^{64}\text{ZnCH}_3$ showing the observed K components of the $J = 116 \leftarrow 115$ transition near 273 GHz. The relative frequency spacing of the components (see text), shown below the spectrum, indicates that the molecule is clearly a prolate symmetric top. The 110 MHz wide spectrum was acquired in ~ 70 s. Asterisks mark unidentified lines arising from other species.

K components displayed in the figure is distinct to a prolate symmetric top. For this type of molecule, the rotational energy levels follow the expression³³

$$F(J, K) = BJ(J+1) - D_J(J(J+1))^2 + (A-B)K^2 - D_{JK}J(J+1)K^2 + H_{JK}J^2(J+1)^2K^2 \quad (1)$$

Here J is the rotational quantum number and K is a quantum number related to the projection of the rotational angular momentum onto the C_3 axis ($\text{I}-\text{Zn}-\text{C}$ axis). A and B are rotational constants, while D_J , D_{JK} (fourth order), and H_{JK} (sixth order) are centrifugal distortion terms. The most intense rotational transitions are governed by the electric dipole selection rules $\Delta J = \pm 1$ and $\Delta K = 0$. Therefore, a symmetric top spectral pattern is immediately recognizable by the presence of K components within a rotational transition ($J+1 \leftarrow J$) having a relative frequency spacing of 1:3:5:7... (due to the $-2D_{JK}(J+1)K^2$ term in the transition frequency expression) starting with $K=0$, as labeled in the spectrum in Figure 1. Note that the $K=3$ and 6 components exhibit approximately a factor of 2 greater intensity relative to adjacent K components—a clear departure from a Boltzmann distribution. This effect results from Fermi–Dirac nuclear spin statistics due to exchange of the three identical protons ($I=1/2$) of the methyl group on rotation of the molecule about

the C_3 axis.³³ By comparing relative intensities of the K components, the rotational temperature is estimated to be ~ 500 K.

Representative spectra of two IZnCH_3 isotopologues are presented in Figure 2. The $K=0-6$ components of the $J=$

IZnCH₃ Isotopologues: K Components

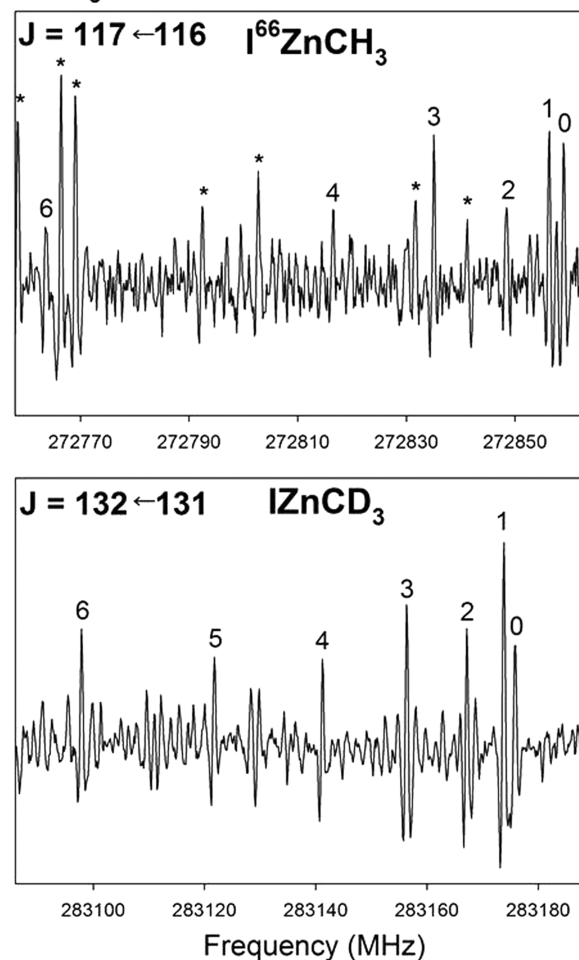


Figure 2. Rotational spectra of $\text{I}^{66}\text{ZnCH}_3$ ($J = 117 \leftarrow 116$ transition, top) and $\text{I}^{64}\text{ZnCD}_3$ ($J = 132 \leftarrow 131$ transition, bottom) recorded near 273 and 283 GHz, respectively. K components are labeled for each species. Contaminant lines are labeled with an asterisk, including one that obscures the $K=5$ component in $\text{I}^{66}\text{ZnCH}_3$. These spectra were generated by averaging 3 and 10 110 MHz-wide scans, respectively; each scan was ~ 70 s in duration.

$117 \leftarrow 116$ rotational transition of $\text{I}^{66}\text{ZnCH}_3$ are displayed in the upper panel, while in the lower panel the K components of the $J = 132 \leftarrow 131$ transition of $\text{I}^{64}\text{ZnCD}_3$ are shown. The $K=5$ component of $\text{I}^{66}\text{ZnCH}_3$ is obscured by a contaminant line, marked by an asterisk. The lines of $\text{I}^{66}\text{ZnCH}_3$ are weaker than those of $\text{I}^{64}\text{ZnCH}_3$ (see Figure 1) by a factor of about 2, reflecting the natural zinc abundance ratio. The increased intensity of the $K=3$ and 6 components (relative to adjacent K components) in both species is consistent with Fermi–Dirac nuclear spin statistics ($\text{I}^{66}\text{ZnCH}_3$) and Bose–Einstein nuclear spin statistics ($\text{I}^{64}\text{ZnCD}_3$, expected factor of intensity increase is 1.4).³³

In addition, another symmetric top pattern was found in the data that resembled that of $\text{I}^{64}\text{ZnCH}_3$ but was significantly

Table 1. Selected Rotational Transition Frequencies of Isotopologues of IZnCH_3 ($\tilde{X}^1\text{A}_1$)^a

$J' \leftarrow J''$	K	$\text{I}^{64}\text{ZnCH}_3$ ($\nu = 0$)		$\text{I}^{66}\text{ZnCH}_3$		$\text{I}^{64}\text{ZnCD}_3$		$\text{I}^{64}\text{Zn}^{13}\text{CH}_3$	
		ν_{obs}	$\nu_{\text{obs-calc}}$	ν_{obs}	$\nu_{\text{obs-calc}}$	ν_{obs}	$\nu_{\text{obs-calc}}$	ν_{obs}	$\nu_{\text{obs-calc}}$
116 $\leftarrow$ 115	0	272 768.911	0.006	270 538.979	0.001				
116 $\leftarrow$ 115	1	272 766.247	0.002	270 536.329	−0.019				
116 $\leftarrow$ 115	2	272 758.280	0.016	270 528.481	0.023				
116 $\leftarrow$ 115	3	272 744.986	0.023	270 515.312	0.004				
116 $\leftarrow$ 115	4	272 726.325	−0.016	270 496.928	0.031				
116 $\leftarrow$ 115	5	272 702.394	−0.005	270 473.192	−0.035				
116 $\leftarrow$ 115	6	272 673.133	−0.003	270 444.315	0.018				
117 $\leftarrow$ 116	0	275 107.700	−0.034	272 858.747	0.012				
117 $\leftarrow$ 116	1	275 105.061	0.009	272 856.068	−0.016				
117 $\leftarrow$ 116	2	275 097.015	0.008	272 848.178	0.049				
117 $\leftarrow$ 116	3	275 083.620	0.022	272 834.869	−0.002				
117 $\leftarrow$ 116	4	275 064.809	−0.017	272 816.279	−0.031				
117 $\leftarrow$ 116	5	275 040.703	0.012						
117 $\leftarrow$ 116	6	275 011.213	0.021	272 763.317	0.039				
125 $\leftarrow$ 124	0					268 240.594	0.007	286 748.130	0.009
125 $\leftarrow$ 124	1					268 238.549	0.020	286 745.381	−0.021
125 $\leftarrow$ 124	2					268 232.368	0.014	286 737.228	−0.016
125 $\leftarrow$ 124	3					268 222.109	0.045	286 723.633	−0.014
125 $\leftarrow$ 124	4					268 207.688	0.030		
125 $\leftarrow$ 124	5								
125 $\leftarrow$ 124	6					268 166.508	0.010		
126 $\leftarrow$ 125	0							289 028.323	0.011
126 $\leftarrow$ 125	1					270 372.945	0.015	289 025.569	−0.002
126 $\leftarrow$ 125	2					270 366.742	0.032	289 017.428	0.081
126 $\leftarrow$ 125	3					270 356.339	−0.006	289 003.644	0.002
126 $\leftarrow$ 125	4								
126 $\leftarrow$ 125	5								
126 $\leftarrow$ 125	6					270 300.357	−0.012		

^aIn MHz.Table 2. Rotational Constants of Isotopologues of IZnCH_3 ($\tilde{X}^1\text{A}_1$)^a

	$\text{I}^{64}\text{ZnCH}_3$ ($\nu = 0$)	$\text{I}^{64}\text{ZnCH}_3$ ($\nu_4 = 1$)	$\text{I}^{66}\text{ZnCH}_3$	$\text{I}^{64}\text{ZnCD}_3$	$\text{I}^{64}\text{Zn}^{13}\text{CH}_3$
B	1178.8444(10)	1176.03988(98)	1169.1951(24)	1075.8051(17)	1150.4001(16)
D_J	0.000115797(40)	0.000115829(34)	0.000114404(88)	0.000090970(51)	0.000109043(51)
D_{JK}	0.011823(57)	0.011746(55)	0.01161(11)	0.008554(93)	0.010877(24)
H_{JK}	$13.2(2.2) \times 10^{-9}$	$12.0(1.9) \times 10^{-9}$	$10.2(4.2) \times 10^{-9}$	$10.3(2.8) \times 10^{-9}$	
rms	0.023	0.028	0.025	0.032	0.035

^aIn MHz; errors are 3σ .

weaker. On the basis of the DFT calculations, these spectra likely arise from rotational transitions within the singly degenerate ($\nu_4 = 1$) I–Zn stretching vibrational mode of $\text{I}^{64}\text{ZnCH}_3$, calculated to be $\sim 230 \text{ cm}^{-1}$ or $\sim 330 \text{ K}$ above the ground state (see Supporting Information Table 2). Although the ($\nu_8 = 1$) I–Zn–C bending mode lies lower in energy ($\sim 100 \text{ cm}^{-1}$), it is doubly degenerate, and subsequently, rotational transitions within this mode would have a more complex pattern. Considering the use of the Broida oven in the molecular synthesis, rotational energy levels in low-energy vibrationally excited states may be populated. From the spectra, the estimated vibrational temperature is $\sim 500 \text{ K}$, which would allow for the rotational energy levels of the ν_4 mode to be populated. The rotational and vibrational degrees of freedom therefore appear to be roughly equilibrated.

Nine rotational transitions were measured for $\text{I}^{64}\text{ZnCH}_3$ and $\text{I}^{64}\text{ZnCD}_3$, eight for $\text{I}^{64}\text{Zn}^{13}\text{CH}_3$ and $\text{I}^{64}\text{ZnCH}_3$ ($\nu_4 = 1$), and six for $\text{I}^{66}\text{ZnCH}_3$, all in their $\tilde{X}^1\text{A}_1$ states, in the frequency range 256–293 GHz. Table 1 gives a sample of the measured

rotational transitions, while the complete list can be found in Table 1 of the Supporting Information. As shown in the table, $K = 0$ –6 components were typically measured for each rotational transition $J + 1 \leftarrow J$ except for $\text{I}^{64}\text{Zn}^{13}\text{CH}_3$, where only $K = 0$ –3 components were observed. Iodine and deuterium electric quadrupole hyperfine splittings were not resolved, as expected at the high J values of the observed rotational transitions.

The pure rotational transition frequencies for each of the four isotopologues and the $\nu_4 = 1$ state of $\text{I}^{64}\text{ZnCH}_3$ were modeled with a symmetric top Hamiltonian with energy eigenvalues given in eq 1. (We can be confident of the rotational quantum number assignments because changing J by ± 1 from the current values increases the rms of a fit for an individual isotopologue by a factor of 30 or more.) Rotational and centrifugal distortion constants were determined for each species from this Hamiltonian using the nonlinear least-squares fitting routine SYMF.³⁴ The derived molecular parameters are listed in Table 2. In addition to the rotational constant, B , the

Table 3. Molecular Structures of IZnCH_3 ($\tilde{X}^1\text{A}_1$) and Related Species

molecule	r_{MX} (Å)	r_{MC} (Å)	r_{CH} (Å)	$\angle\text{M}-\text{C}-\text{H}$ (deg)	$\angle\text{H}-\text{C}-\text{H}$ (deg)	method	ref
IZnCH_3	2.4076(2)	1.9201(2)	1.105(9)	110.2(5)	108.7(5)	r_0	this work
	2.4373	1.9144	1.0951	111.125	107.825	B3LYP/6-31G*/LanL2DZ	13
	2.4471	1.9393	1.0904	110.192	108.741	B3PW91/6-311G++(3df,2p)/LanL2DZ	this work
	2.4606	1.9518	1.0892	110.041	108.895	B3LYP/6-311G++(3df,2p)/LanL2DZ	this work
	2.4508	1.9383	1.0904	110.201	108.731	B3PW91/6-311G++(3df,2p)/DGDZVP	this work
	2.4689	1.9505	1.0892	110.031	108.906	B3LYP/6-311G++(3df,2p)/DGDZVP	this work
ZnI	2.2914						36
ZnCH_3		2.001 (7)	1.105 ^a		109.21 ^a	r_0	32
HZnCH_3	1.5209(1)	1.9281(2)	1.140(9)	110.2(3)	108.7(3)	r_0	35
$\text{IZnCH}_2\text{CH}_3^b$	2.64						17
CuCH_3		1.8841(2)	1.091(2)		110.07(8)	r_0	31
ICH_3		2.1392	1.113		111.42	r_0	37
CH_4			1.09403(1)		109.5	r_0	38

^aHeld fixed. ^bX-ray crystal structure.

centrifugal distortion constants D_J , D_{JK} , and H_{JK} were necessary for an acceptable fit. Because only the $K = 0-3$ components were observed for $\text{I}^{64}\text{Zn}^{13}\text{CH}_3$, H_{JK} could not be determined for this isotopologue. The rms of the individual fits are in the range 23–35 kHz, less than the estimated experimental uncertainty of ± 50 kHz (see Table 2).

The bond lengths and bond angles of methylzinc iodide in its $\tilde{X}^1\text{A}_1$ electronic ground state were determined from the rotational constants, B , of the four isotopologues, recognizing that the molecule belongs to the C_{3v} point group. A nonlinear least-squares analysis was carried out to determine an r_0 (ground vibrational state averaged) structure. Derived r_0 structural parameters are listed in Table 3; the accuracy of an r_0 structure has been discussed in detail elsewhere.³⁵

DFT values of the geometric parameters for IZnCH_3 are reported in Table 3. Geometric optimizations were also performed for IZnCH_3 with an initial I–Zn–C bond angle of 100° in order to confirm that the structure with C_3 symmetry was the global minimum. For all functional and basis set combinations used, the I–Zn–C bond angle was optimized to 180° . To rule out the possibility of ZnICH_3 as the source of our millimeter-wave spectra, a geometric optimization and harmonic vibrational analysis, with the Zn–I–C bond angle fixed to 180° , was performed on this isomer. For each computational method listed, ZnICH_3 corresponded to a higher energy than IZnCH_3 and can be ruled out as the species responsible for the spectra observed. For the B3LYP functional, vibrational analysis resulted in imaginary vibrational frequencies for ZnICH_3 .

The computationally determined vibrational frequencies and dipole moment for IZnCH_3 are listed in Table 2 of the Supporting Information. From infrared measurements of IZnCH_3 , the Zn–C (ν_3) stretching frequency is estimated to be 530 cm^{-1} in THF and 523 cm^{-1} in DME.¹⁵ The methyl group twist (ν_7) frequency is reported to be 640 cm^{-1} in THF and 654 cm^{-1} in DME.¹⁵ The DFT-calculated frequencies for these vibrational motions are ~ 570 and $\sim 700\text{ cm}^{-1}$, respectively. The experimental and DFT values are in reasonable agreement, considering the solvent effects associated with the IR measurements and that no scaling factors have been applied to the DFT values.

DISCUSSION

The experimentally derived structure for IZnCH_3 is shown in Figure 3, and the parameters are listed in Table 3. Along the C_3

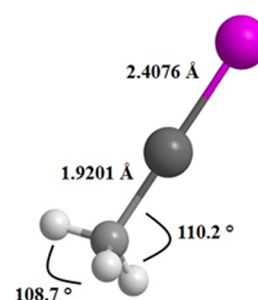


Figure 3. Derived experimental structure for IZnCH_3 in its $\tilde{X}^1\text{A}_1$ ground electronic state.

axis of symmetry, the zinc–iodine and zinc–carbon bond lengths were determined to be $r_{\text{Zn-I}} = 2.4076(2)\text{ Å}$ and $r_{\text{Zn-C}} = 1.9201(2)\text{ Å}$. For the methyl group, the C–H bond distance is $r_{\text{C-H}} = 1.105(9)\text{ Å}$ with a H–C–H bond angle of $108.7(5)^\circ$.

Computational values of the geometric parameters are also reported in Table 3 and compare reasonably well to the corresponding experimental r_0 values. As mentioned, all functional and basis set combinations optimized to the geometry with a C_3 axis of symmetry, even when initially starting with a bent I–Zn–C backbone of 100° . Furthermore, the computational work ruled out the ZnICH_3 isomer. Little difference is noted in the geometrical parameters obtained using either the B3LYP or the B3PW91 functional (see Table 3); both have been used in previous calculations involving RZnX species.^{17–19} Structural parameters reported in ref 13 using the B3LYP functional, the 6-31G* basis set for H, C, and Zn, and the LanL2DZ basis set and ECP for iodine compare well to our computational values; the slight differences are due to our choice of a larger basis set, 6-311G++(3df,2p), for H, C, and Zn.

Geometries for related species are also listed in Table 3. Surprisingly, in a comparison of IZnCH_3 with methane, substitution of an H atom with the I–Zn moiety does not change the C–H bond length by more than $\sim 0.01\text{ Å}$ and the H–C–H angle by less than 1° . On the other hand, there is a notable increase in the Zn–I bond length of $>0.1\text{ Å}$ relative to the diatomic species ZnI (2.4076 vs 2.2914 Å). The Zn–I bond length elongates even more in $\text{IZnCH}_2\text{CH}_3$, as indicated by the crystal structure (2.64 Å ; see Table 3). Diatomic ZnI is likely to be quite ionic. The increased Zn–I bond distance suggests that the Zn–I bond character is different in IZnCH_3 and

IZnCH₂CH₃, most likely less ionic. Furthermore, the Zn–C–H (or H–C–H) bond angle and Zn–C bond length appear to be virtually identical in both HZnCH₃ and IZnCH₃. Therefore, the basic molecular structure is not significantly altered by substitution of H with iodine. The zinc bonds in HZnCH₃ have a significant covalent character based on the quadrupole coupling constant of H⁶⁷ZnCH₃ (ref 35) with the 4s and 4p orbitals on zinc likely sp hybridized as proposed by Castillo et al.³⁹ The similarity of the IZnCH₃ structure to that of HZnCH₃ suggests that the zinc bonds in IZnCH₃ have substantial covalent character as well.

The H–C–H methyl group angle is slightly smaller in both IZnCH₃ and HZnCH₃ than in methane, as shown in Table 3. Steric hindrance may be slightly closing the H–C–H angle in the zinc species. Curiously, substitution of one hydrogen in methane with iodine or copper appears to slightly open this angle. It is not clear why this effect occurs.

CH₃I has been used as a satisfactory methyl group donor in the gas-phase synthesis of alkali, alkaline-earth, and copper monomethyl species.^{29–31} Therefore, it was quite surprising that we were able to produce IZnCH₃ by reaction of zinc metal and CH₃I in these experiments. Spectral lines of HZnCH₃, ZnCH₃, and ZnI were not apparent in the data. In the case of HZnCH₃, the frequency region of one of the measured rotational transitions from Flory et al.³⁵ was scanned with considerable signal averaging but no lines were observed. Although the pure rotational spectra of ZnCH₃ and ZnI have not yet been measured, there are good estimates of their rotational constants from optical data,^{32,36} and the fine structure splitting due to their doublet electronic ground states would allow for their obvious recognition. If the recombination of radical fragments created in the dc discharge was the predominant mechanism for synthesis of IZnCH₃ then it is reasonable to also expect formation of ZnCH₃, ZnI, and HZnCH₃. Not observing these species in the discharge mixture may suggest an alternative route; however, additional experimental studies would be needed to confirm this possibility.

Oxidative addition of Zn to the C–H bond of methane and the C–Cl bond of CH₃Cl has been examined using DFT and the Activation Strain model by de Jong et al.⁴⁰ These calculations predict that the insertion occurs through a transition state with C_s symmetry, where the C–Cl or C–H bond elongates and the C, Zn, and Cl (or H) atoms form a triangle in a plane that bisects one H–C–H group. Activation energies for zinc insertion into the C–H bond and the C–Cl bond were found to be 91.6 and 44.3 kcal/mol, respectively. The energy barrier difference reflects the higher bond strength or enthalpy⁴¹ of the C–H unit (397 kJ/mol) relative to the C–Cl bond (328 kJ/mol). The C–I bond has a bond enthalpy of 220 kJ/mol⁴²—smaller than that of both C–Cl and C–H bonds. Therefore, an even lower activation energy for zinc insertion into the C–I bond of CH₃I might be expected. Our previous studies have shown that HZnCH₃ can be formed via a zinc insertion into methane using the same gas-phase experimental methods as in this work.³⁵ The predictions of de Jong et al. suggest oxidative addition of zinc to CH₃I as a highly likely gas-phase synthetic route to IZnCH₃.

Additional evidence for formation of IZnCH₃ through direct insertion of zinc into the I–C bond of CH₃I arises from the required use of a dc discharge in the molecular synthesis. The pale purple color of the plasma in the dc discharge used to create IZnCH₃ has been observed in the synthesis of ZnCCH,

HZnCH₃, and other zinc-bearing radicals.^{35,43} This has been attributed to emission from the ¹P and/or ³P excited states of atomic zinc.^{39,44} It has been speculated that zinc may have to be in an excited state in order to insert into the C–H bond of methane,⁴⁵ and this may also be the case for the C–I bond. Insertion of activated zinc into the carbon–iodine bond would require the presence of a third body for stabilization, which is provided by the argon carrier gas.

CONCLUSIONS

The geometry of IZnCH₃ has been determined using gas-phase techniques of millimeter-wave rotational spectroscopy—the first experimental structure of a monomeric alkylmetal halide. This model system was found to have a ¹A₁ ground electronic state and belong to the C_{3v} point group. Spectroscopic measurements suggest that IZnCH₃ is formed by facile insertion of zinc into the carbon–iodine bond of CH₃I. The derived structure and formation mechanism found in this work are consistent with DFT computational studies of organozinc halide reagents.

ASSOCIATED CONTENT

Supporting Information

Complete list of rotational transition frequencies for all isotopologues of IZnCH₃ and DFT-calculated vibrational frequencies and dipole moments. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This research was supported by NSF Grant CHE 10-57924. J.P.Y. and P.M.S. thank Canisius College for providing travel funds and the Ziurys group for their hospitality.

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